KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 3 REPORT

Cleaning Sprint

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
Atchaya S	25BAD016

1. Week Contribution:

Studied the effect of missing values and outliers on statistical analysis, reviewed the cleaning workflow, and maintained the related files in GitHub.

2. Work Done:

During Week 3, my individual work focused on studied the effect of missing values and outliers on statistical analysis, reviewed the cleaning workflow, and maintained the related files in GitHub.

- Reviewed the missing-value condition in the selected dataset and considered how incomplete records can affect statistical summaries and visualisations.
- Studied suitable approaches for handling missing values and the importance of checking the dataset before analysis.
- Explored outlier detection concepts, including the Interquartile Range (IQR) and Z-score methods.
- Reviewed the cleaned dataset and the data-quality checks used in the project, including the final missing-value check.
- Maintained the Week 3 cleaning documentation/notebook and the relevant raw and cleaned dataset versions in the GitHub repository.

3. Topics Covered:

- Missing-value identification and its impact on analysis.
- Outlier detection using IQR and Z-score concepts.
- Data-quality checking and validation.
- Comparison of raw and cleaned data.
- GitHub management of dataset versions and cleaning-stage files.

4. Outcome of My Contribution:

This stage improved my understanding of why data quality is important before statistical analysis. Maintaining both raw and cleaned dataset files in GitHub helped keep the project workflow organised and traceable.

5. GitHub Repository:

Repository: https://github.com/atchayas-ai/DSV_Team_16_Youth_Smoking_Drug_Abuse

I helped manage and maintain the shared repository by keeping the relevant weekly documentation, notebooks, datasets and project materials organised. The exact files updated during each week are described in the Work Done section.

6. Reference:

Project repository: GitHub – Team 16, Youth Smoking and Drug Consumption Analysis.